

MUHAMMAD BASITH K

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SUMMARY

Software Developer experienced in Python, C++, machine learning, and product-focused engineering. Skilled in building ML applications, geospatial analytics systems, and API-driven solutions. Strong foundation in DSA, problem-solving, and creating functional prototypes.

SKILLS

Languages: Python, C++, SQL

Frameworks & Libraries: PyTorch, Scikit-Learn, OpenCV, Pandas, NumPy

Tools & Platforms: GitHub, Jupyter, Unreal Engine, Plotly, REST APIs

Core Concepts: OOP, DSA, ML Pipelines, Neural Networks, Computer Vision

Practices: Agile, Debugging, Version Control, Prototyping

PROJECTS

TERRAWATCH – ML-Based Disaster Prediction Platform (2025)

- Built a geospatial analytics platform integrating ML models with 3D visualization for disaster susceptibility analysis.
- Processed 6 raster datasets at 30m resolution, improving valid-data coverage to ~90%.
- Applied clustering and neural models to generate predictive risk maps.
- Developed an interactive prototype simulating real-world early-warning workflows.

Emotion Detection & Music Recommendation (2025)

- Built a real-time emotion recognition system using FER-2013 with CNN-based architecture.
- Implemented full ML pipeline: face detection → emotion inference → Spotify music recommendation.
- Integrated PyTorch, OpenCV, and Spotify API for real-time output.

CERTIFICATIONS

- Fortinet Certified Associate in Cybersecurity
- AWS Academy Graduate – Machine Learning Foundations
- Cisco Networking Essentials (Cisco)
- Oracle Certified Foundations Associate (Oracle)

EDUCATION

B.Tech in Computer Science Engineering

SRM Institute of Science and Technology, Chennai (2022–Present) | CGPA: 7.13

Higher Secondary (CS & Math)

NM HSS Thirunavaya | 1041/1200

CBSE Class 10

MES School Kuttipuram | 369/500

ACHIEVEMENTS & RESPONSIBILITIES

- Developed multiple ML prototypes integrating real-time data and cloud concepts.
- Led a 3-member team for TerraWatch ML model and system design.
- Active contributor to ML and software projects on GitHub.